# Maxwell Rosen

## Undergraduate | UC San Diego

Southern California
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## Personal Statement:

I'm pursuing a Chemical Engineering major with a Bioengineering focus alongside a Business Economics minor, driven by a long-term goal of engineering medical devices.

## Education

### **University of California, San Diego, CA** - *Bachelor of Science*

September 2024 - June 2028

- Chemical Engineering Major | Business Economics Minor
- GPA: 3.96
- Relevant Coursework: Thermodynamics, Material and Energy Balances, Organic Chemistry, Organizational Leadership

### **La Cañada High School** - *High School Diploma*

August 2020 - June 2024

- GPA: 4.00
- Relevant Coursework: Sports Medicine, Medical Biology
- AP Scores: 5 Calculus BC, 5 Chemistry, 5 Biology, 4 AP Physics, 5 European History
- Pasadena City College Dual Enrollment: Multivariable Calculus, Career Planning, Linear Algebra

## Skills

- Python, Raspberry Pi, Github, MATLAB, CAD Autodesk Fusion, Microsoft Excel

## Experience

### **Microfluidic Device Research** - *Visiting Undergraduate Researcher*

National Taiwan University, Taipei, Taiwan • June 2026 - July 2026

- Selected for competitive UCEAP summer research at National Taiwan University to design, fabricate, and characterize microfluidic devices for biomedical applications
- Calculating saturation levels of a membrane by mass and imaging to optimize Western Blot experiments
- Developing an automatic micropipetting module for Western Blot and IHC

**Salk Institute Research Intern** - *Lab Intern*

Salk Institute, Ecker Lab • September 2025 - August 2026

- Designed and deployed 4 Raspberry Pi–based high-throughput imaging systems at 1% the cost of commercial alternatives ($500 vs. $50,000 per unit) to monitor plant growth across 200+ samples under variable environmental stress
- Collaborated on a 4-person team to troubleshoot and solve hardware-software integration challenges (image pipeline scaling, GPIO obstacles) while fulfilling time-sensitive project parameters
- Documented system architecture and authored setup protocols

**LongevityBench: LLM Benchmarking for Aging Biology Hackathon** - *Hackathon Team Member*

Caltech Longevity Hackathon, Pasadena, California • May 23-24, 2026

- Worked in a team to build LongevityBench, a benchmark suite evaluating large language models on reasoning and trustworthiness in aging biology, submitted to Track 1 (LongevityLLM Benchmarking) sponsored by Insilico Medicine
- Sourced and validated the domain-specific datasets (Lipidomics, Gut Microbiomes) used to construct the evaluation tasks, and supported task design and analysis benchmarking a fine-tuned longevity language model

**Head Counselor & Inclusion Specialist** - *MESSH Support, Inclusion/General Counselor*

Shalom Institute • Summers & Year Round 2023 - 2025

- Led a cabin of 10+ campers per cabin while delivering one-on-one support for the Mental, Emotional, Social, and Spiritual Health (MESSH) of 5 campers with special needs
- Designed and facilitated 20 programs per summer to promote values like building community, environmental stewardship, and role modeling

**Lighting Technician** - *Utility*

Showpro, Freelance • September 2025 - Present

- Execute live event production through fixture placement, cable routing, dimmer/DMX setup, and power distribution to meet show specifications and ensure efficient run-of-show operations

**Triton Engineering Student Council** - *Operations Committee*

UC San Diego • March 2025 - June 2025

- Organized professional development and networking events for engineering students
- Strengthened project coordination, communication, and logistical planning

**Goldman Sachs Possibilities Summit** - *Engineering Summit*

Goldman Sachs • January 2025 - June 2025

- Cultivated top industry skills through networking events and webinars

**Athletic Trainer Internship** - *Senior Intern*

LCHS • April 2022 - June 2024

- Shadowed licensed Athletic Trainer in injury prevention and rehabilitation
- Administered first aid and injury prevention to 40+ student-athletes
- Mentored 4 junior interns, developed hands-on care, attention to detail, problem solving under pressure, and communication skills

### **Science National Honors Society** - *Secretary*

LCHS • October 2021 - June 2024

- Organized and volunteered in 15 community science events (20-40 participants), arranged 25 science-related lectures (5-40 student audiences), operated the requirement tracker, and formulated/sent the weekly update to our 120+ High School members
- Collaborated with 4 other board members

### **Technology Advisory Council** - *Student Advisor*

LCHS • November 2020 - June 2023

- Developed improvements to LCHS's communication and efficiency between students and staff through technology

### **Wrestling** - *Varsity Starter*

LCHS • July 2022 - June 2024

- Practiced with and mentored 40 athletes
- Wrestled in 23 regional and national competitions
- CIF Champion for Character (see Awards)

## Awards

### CIF Champion for Character

September 2023

- 1 of 13 student-athletes, chosen from thousands of applicants in the CIF Southern Section
- Honored my commitment to leadership in upholding the CIF ideals of sportsmanship, integrity, positive relations, respect for diversity of all people, competition, fair play over victory, and scrupulous adherence to the rules of Wrestling

### Commercial Photography Awards

March 2024

- Award of Excellence & Honorable Mentions by Working Assumptions in WorkXFamily
- Honorable Mention in California Streaming 2024 Digital Photo Contest

## Volunteer Experience

### AIChe National Conference - *10/27/2024*

5.5 Hours • 9 AM - 12 PM & 5:30 PM - 8 PM

- Assisted checking in for K-12 showcase event, and assisted badge check-in for general event
- Supported 30 colleges and 200+ chemical engineers from around the world at this event

### Red Cross Club Blood Drive - *3/1/2023, 11/6/2023*

12 Hours

- Assisted with Set-Up, Check-in, and Take-Down
- Donated blood as a participant

### Operation Gratitude, Los Angeles, CA - *2023*

6 Hours

- Boxed 30 care packages (personal hygiene, encouraging notes, snacks, etc.) for American soldiers on the front lines

### Math Competition Volunteer - *2022-2024*

25 Hours

- Assisted with Set-Up, Check-In, Grading, and Takedown
- Events: 3x Mathzilla (3 Elementary Schools), 2x Aleph (Middle Schools from around Los Angeles area)